KAIZEN MATHS

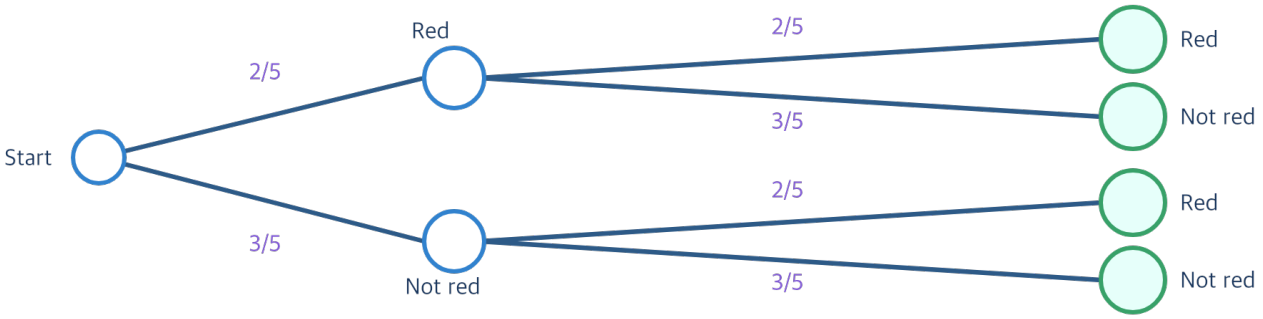
Lesson starter

Tree Diagrams and Conditional Probability

Independent Events

Name _____

SA1



A spinner lands on red has probability $\frac{2}{5}$ each time. The event is repeated twice independently. Find P(Red then Red).

STA-3EBB-L1 · SA

Show your working

KAIZEN MATHS

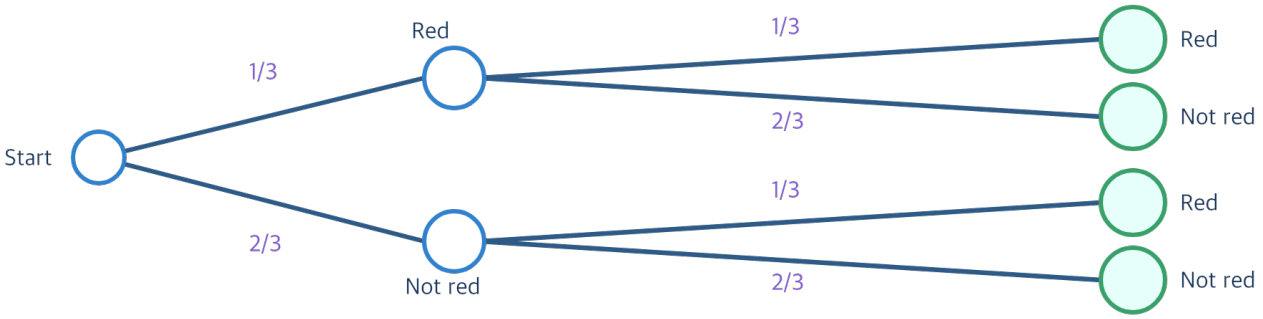
Lesson starter

Tree Diagrams and Conditional Probability

Independent Events

Name _____

SB1



A spinner lands on red has probability $\frac{1}{3}$ each time. The event is repeated twice independently. Find the probability of exactly one red.

STA-3EBB-L1 · SB

Show your working

KAIZEN MATHS

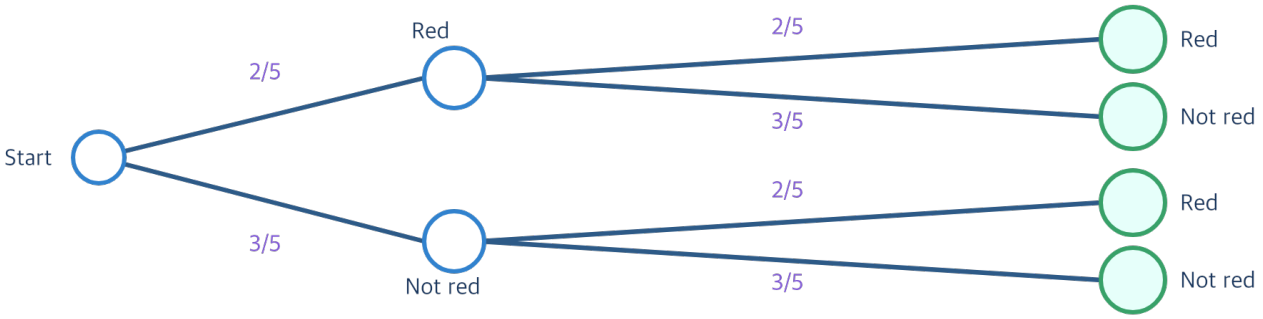
Lesson starter

Tree Diagrams and Conditional Probability

Independent Events

Name _____

SC1



A spinner lands on red has probability $\frac{2}{5}$ each time. The event is repeated twice independently. Find the probability of at least one red.

STA-3EBB-L1 · SC

Show your working

KAIZEN MATHS

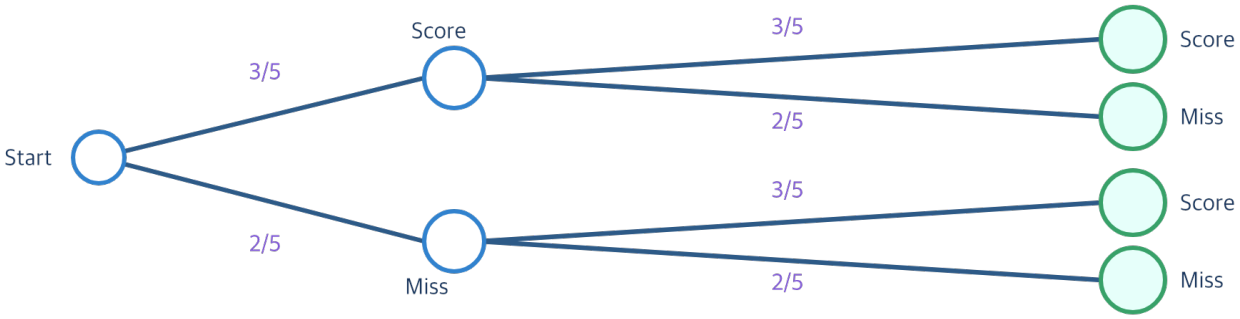
Lesson starter

Tree Diagrams and Conditional Probability

Independent Events

Name _____

SD1



A free throw is scored has probability $\frac{3}{5}$ each time. The event is repeated twice independently. Find P(Score then Score).

STA-3EBB-L1 · SD

Show your working

KAIZEN MATHS

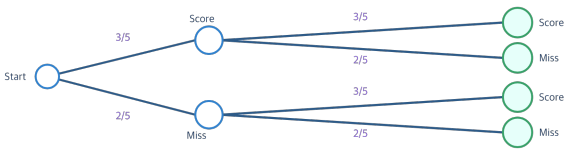
Exit ticket

Tree Diagrams and Conditional Probability

Independent Events

Name _____

EA1



A free throw is scored has probability $\frac{3}{5}$ each time. The event is repeated twice independently. Find the probability of exactly one score.

STA-3EBB-L1 · EA

Show your working

KAIZEN MATHS

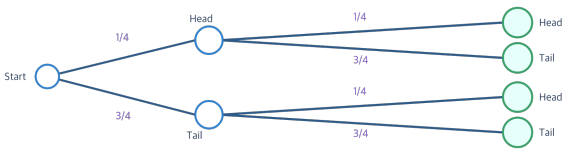
Exit ticket

Tree Diagrams and Conditional Probability

Independent Events

Name _____

EB1



A biased coin lands on heads has probability $\frac{1}{4}$ each time. The event is repeated twice independently. Find the probability of at least one head.

STA-3EBB-L1 · EB

Show your working

KAIZEN MATHS

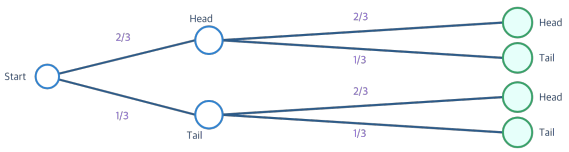
Exit ticket

Tree Diagrams and Conditional Probability

Independent Events

Name _____

EC1



A biased coin lands on heads has probability $\frac{2}{3}$ each time. The event is repeated twice independently. Find P(Head then Head).

STA-3EBB-L1 · EC

Show your working

KAIZEN MATHS

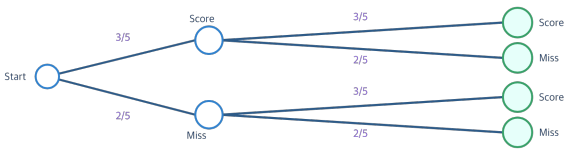
Exit ticket

Tree Diagrams and Conditional Probability

Independent Events

Name _____

EA1



A free throw is scored has probability $\frac{3}{5}$ each time. The event is repeated twice independently. Find the probability of exactly one score.

STA-3EBB-L1 · EA

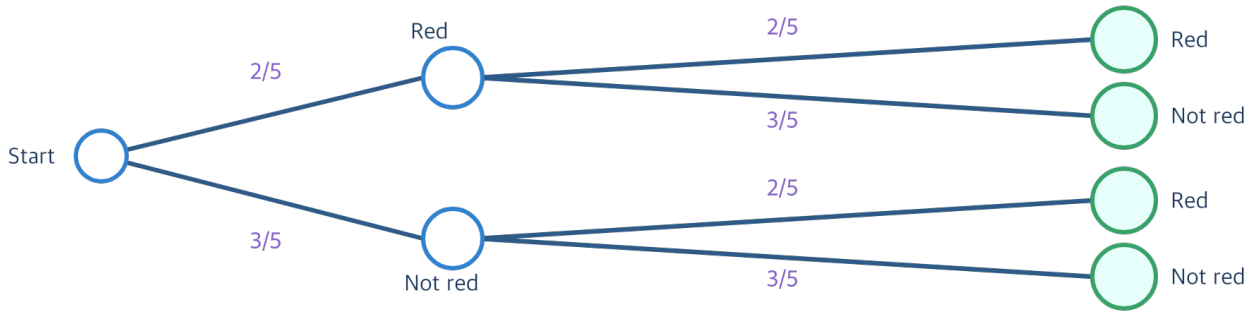
Show your working

Teacher answers & working

Tree Diagrams and Conditional Probability · Independent Events

STA-3EBB-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



A spinner lands on red has probability $\frac{2}{5}$ each time. The event is repeated twice independently. Find P(Red then Red).

Answer:

$\frac{4}{25}$

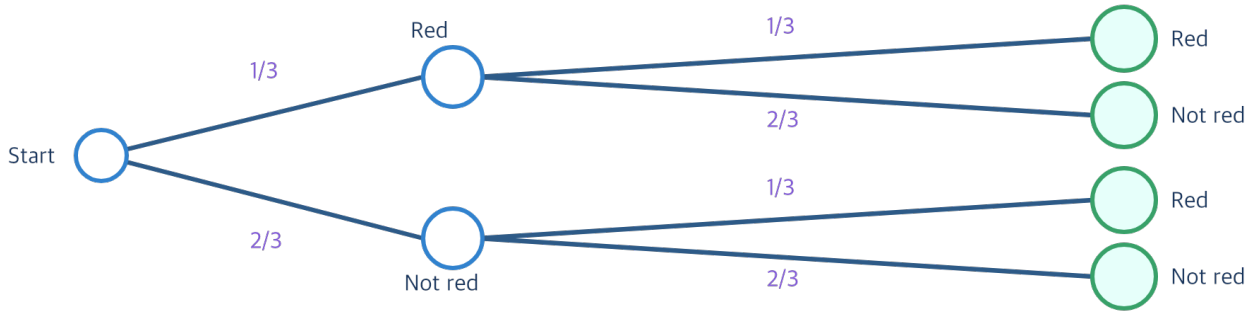
Working:

For independent events, the second-stage probabilities stay the same.

P(Red then Red) = $\frac{2}{5} \times \frac{2}{5}$

$\frac{4}{25} = 16.0\%$

SB1



A spinner lands on red has probability $\frac{1}{3}$ each time. The event is repeated twice independently. Find the probability of exactly one red.

Answer:

$\frac{4}{9}$

Working:

Exactly one success can happen in two separate paths.

Red then Not red: $\frac{1}{3} \times \frac{2}{3} = \frac{2}{9}$

Not red then Red: $\frac{2}{3} \times \frac{1}{3} = \frac{2}{9}$

Add the paths: $\frac{2}{9} + \frac{2}{9} = \frac{4}{9}$

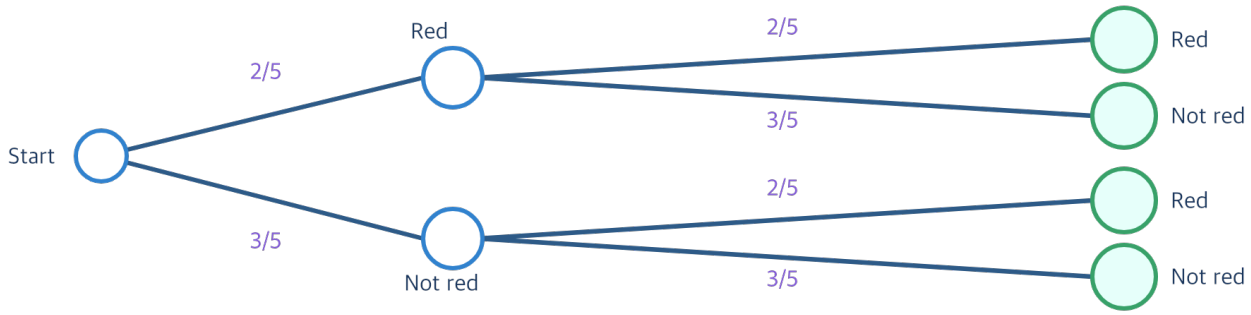
$\frac{4}{9} = 44.4\%$

Teacher answers & working

Tree Diagrams and Conditional Probability · Independent Events

STA-3EBB-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



A spinner lands on red has probability $\frac{2}{5}$ each time. The event is repeated twice independently. Find the probability of at least one red.

Answer:

$\frac{16}{25}$

Working:

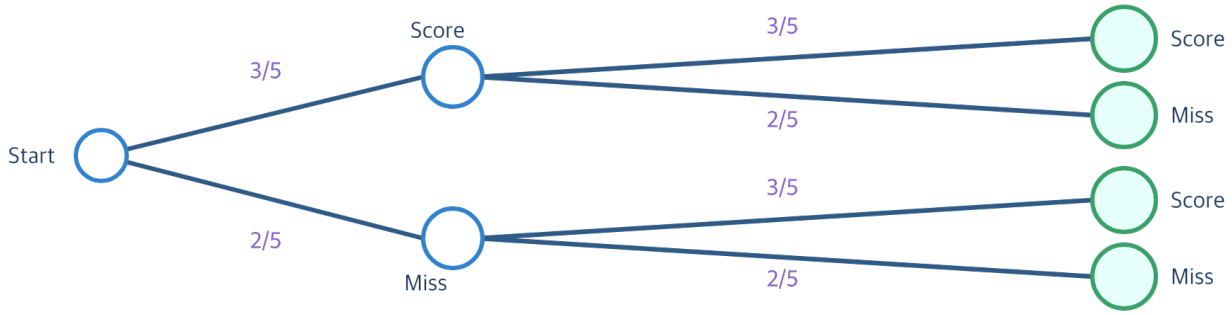
Use the complement: at least one = 1 - none.

P(no red both times) = $\frac{3}{5} \times \frac{3}{5} = \frac{9}{25}$

P(at least one red) = 1 - $\frac{9}{25} = \frac{16}{25}$

$\frac{16}{25} = 64.0\%$

SD1



A free throw is scored has probability $\frac{3}{5}$ each time. The event is repeated twice independently. Find P(Score then Score).

Answer:

$\frac{9}{25}$

Working:

For independent events, the second-stage probabilities stay the same.

P(Score then Score) = $\frac{3}{5} \times \frac{3}{5}$

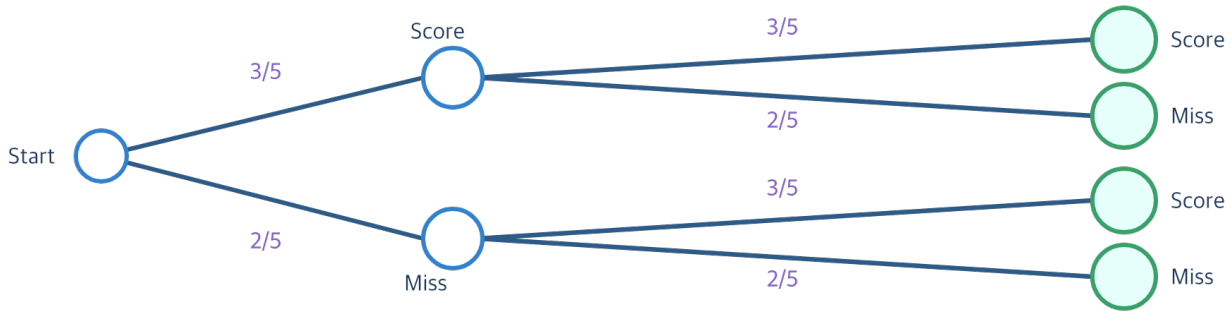
$\frac{9}{25} = 36.0\%$

Teacher answers & working

Tree Diagrams and Conditional Probability · Independent Events

STA-3EBB-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



A free throw is scored has probability $\frac{3}{5}$ each time. The event is repeated twice independently. Find the probability of exactly one score.

Answer:

$\frac{12}{25}$

Working:

Exactly one success can happen in two separate paths.

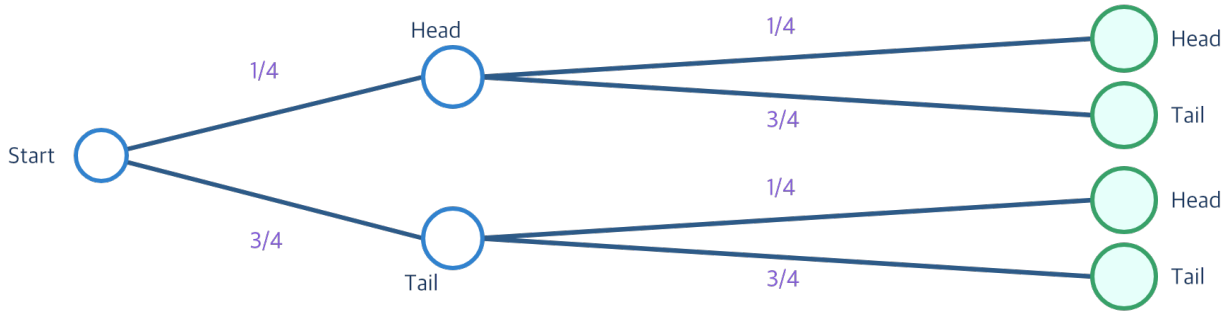
Score then Miss: $\frac{3}{5} \times \frac{2}{5} = \frac{6}{25}$

Miss then Score: $\frac{2}{5} \times \frac{3}{5} = \frac{6}{25}$

Add the paths: $\frac{6}{25} + \frac{6}{25} = \frac{12}{25}$

$\frac{12}{25} = 48.0\%$

EB1



A biased coin lands on heads has probability $\frac{1}{4}$ each time. The event is repeated twice independently. Find the probability of at least one head.

Answer:

$\frac{7}{16}$

Working:

Use the complement: at least one = 1 - none.

P(no head both times) = $\frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$

P(at least one head) = 1 - $\frac{9}{16} = \frac{7}{16}$

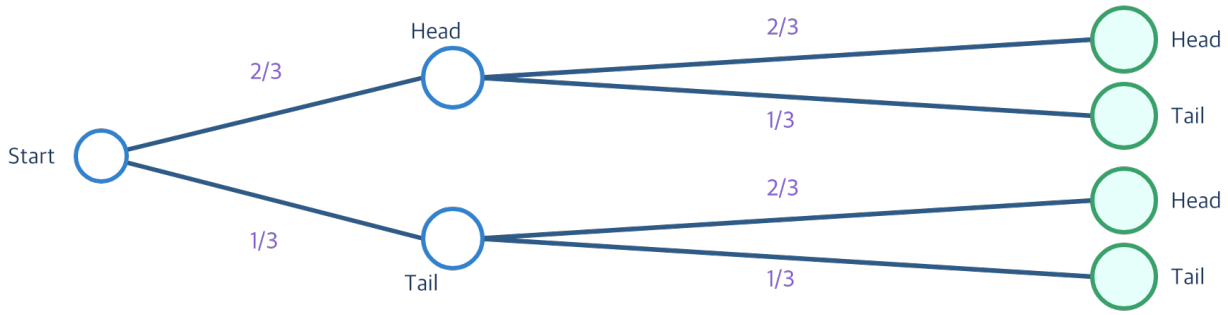
$\frac{7}{16} = 43.8\%$

Teacher answers & working

Tree Diagrams and Conditional Probability · Independent Events

STA-3EBB-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



A biased coin lands on heads has probability $\frac{2}{3}$ each time. The event is repeated twice independently. Find P(Head then Head).

Answer:

$\frac{4}{9}$

Working:

For independent events, the second-stage probabilities stay the same.

P(Head then Head) = $\frac{2}{3} \times \frac{2}{3}$

$\frac{4}{9} = 44.4\%$